



# A Citation-Grounded Benchmark for Trustworthy Earnings Call Transcript Analysis with Large Language Models

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## Motivation

- Financial analysis increasingly requires [claims with verifiable citations](#), not standalone LLM answers.
- Existing evaluation often relies on expensive expert annotation, especially for long earnings call transcripts.
- We distinguish: [groundedness](#) (whether an answer is supported by the source) from [correctness](#) (whether it answers the question correctly).

**Key question:** Can citation-grounded financial analysis be evaluated automatically and at scale?

## Numeric Evidence Evaluation (NEE)

We represent each salient financial figure as

$$F = (s, v, t, m),$$

where  $s$  is the text span,  $v$  is a scalar value or interval,  $t \in \{\text{scalar}, \text{range}\}$ , and  $m$  is the optional unit.

Each figure is normalized into

$$\kappa(F) = (t, \nu(v), \mu(m)),$$

and two figures are considered an exact numeric match iff

$$F \equiv F' \iff \kappa(F) = \kappa(F').$$

For two numeric evidence sets  $X$  and  $Y$ , define

$$X \preceq Y = \{x \in X : \exists y \in Y, x \equiv y\}.$$

NEE measures groundedness at three levels:

$$G_{\text{claim-citation}} = \frac{\sum_i |F_i^{\text{clm}} \preceq F_{i,*}^{\text{cit}}|}{\sum_i |F_i^{\text{clm}}|},$$

$$G_{\text{claim-context}} = \frac{\sum_i |F_i^{\text{clm}} \preceq F^{\text{ctx}}|}{\sum_i |F_i^{\text{clm}}|},$$

$$G_{\text{citation-context}} = \frac{\sum_i \sum_k |F_{i,k}^{\text{cit}} \preceq F^{\text{ctx}}|}{\sum_i \sum_k |F_{i,k}^{\text{cit}}|}.$$

These metrics quantify whether generated claims are supported by their citations, whether claims are supported by the source transcript, and whether citations are faithful to the source transcript.

## Groundedness Benchmark

- We evaluate six GPT-4.1 citation-generation settings: Vanilla, PostCite, and LLM Retrieval (LLMR), each using either citation indices or text.
- All approaches achieve [high claim-context and citation-context groundedness](#). The main difference lies in [claim-citation alignment](#).

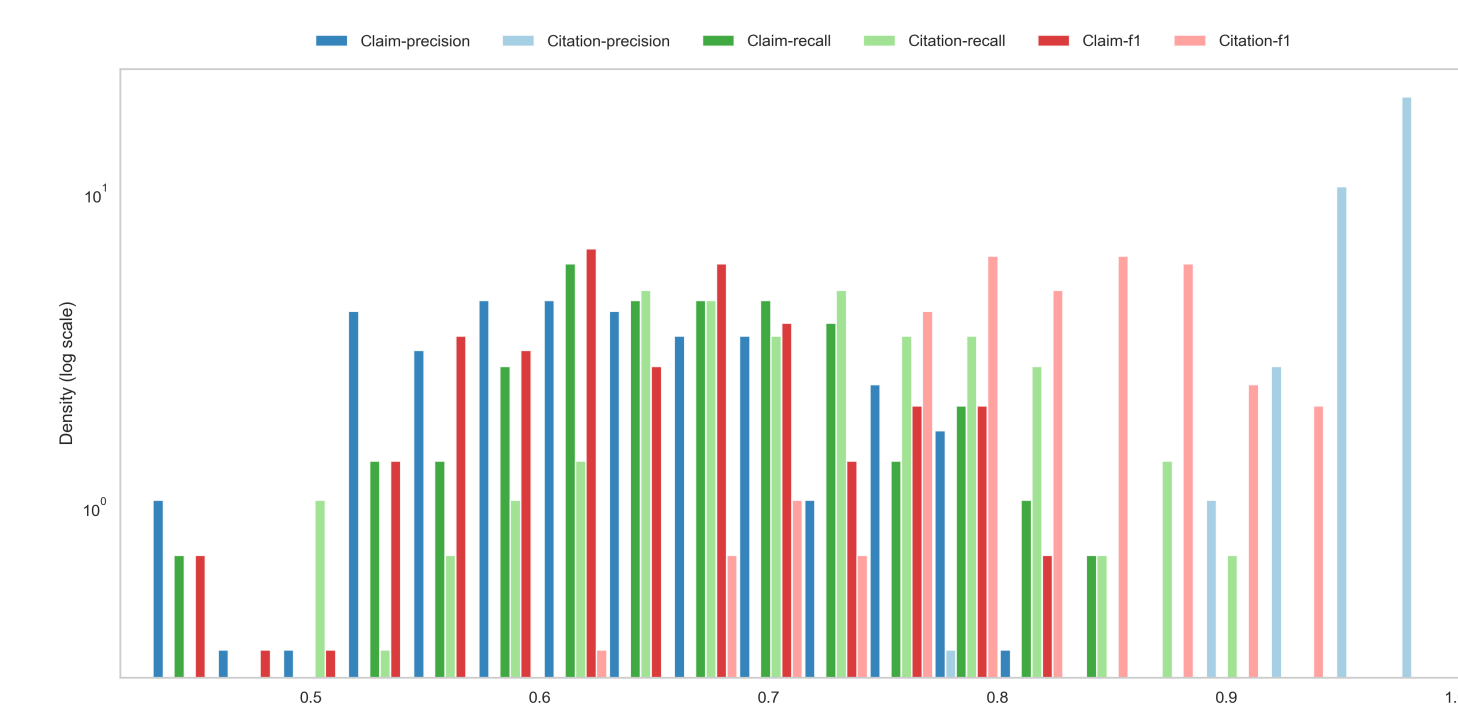
| Metric           | Vanilla |       | PostCite |       | LLMR  |                       |
|------------------|---------|-------|----------|-------|-------|-----------------------|
|                  | Index   | Text  | Index    | Text  | Index | Text                  |
| Claim-Context    | 0.975   | 0.970 | 0.975    | 0.982 | 0.977 | 0.982                 |
| Citation-Context | 0.995   | 0.989 | 0.995    | 0.994 | 0.993 | 0.993                 |
| Claim-Citation   | 0.928   | 0.917 | 0.907    | 0.952 | 0.927 | <a href="#">0.959</a> |

**Table 1:** Groundedness across citation-generation baselines.

## Correctness: ECTs-100

We introduce [ECTs-100](#), an automatically constructed benchmark where salient financial figures define objective question-answer-citation triplets.

- Citations are usually correct when produced: [precision = 0.9721](#).
- Lower citation recall indicates [incomplete evidence coverage](#).
- F1 shows a clear gap between groundedness and correctness.

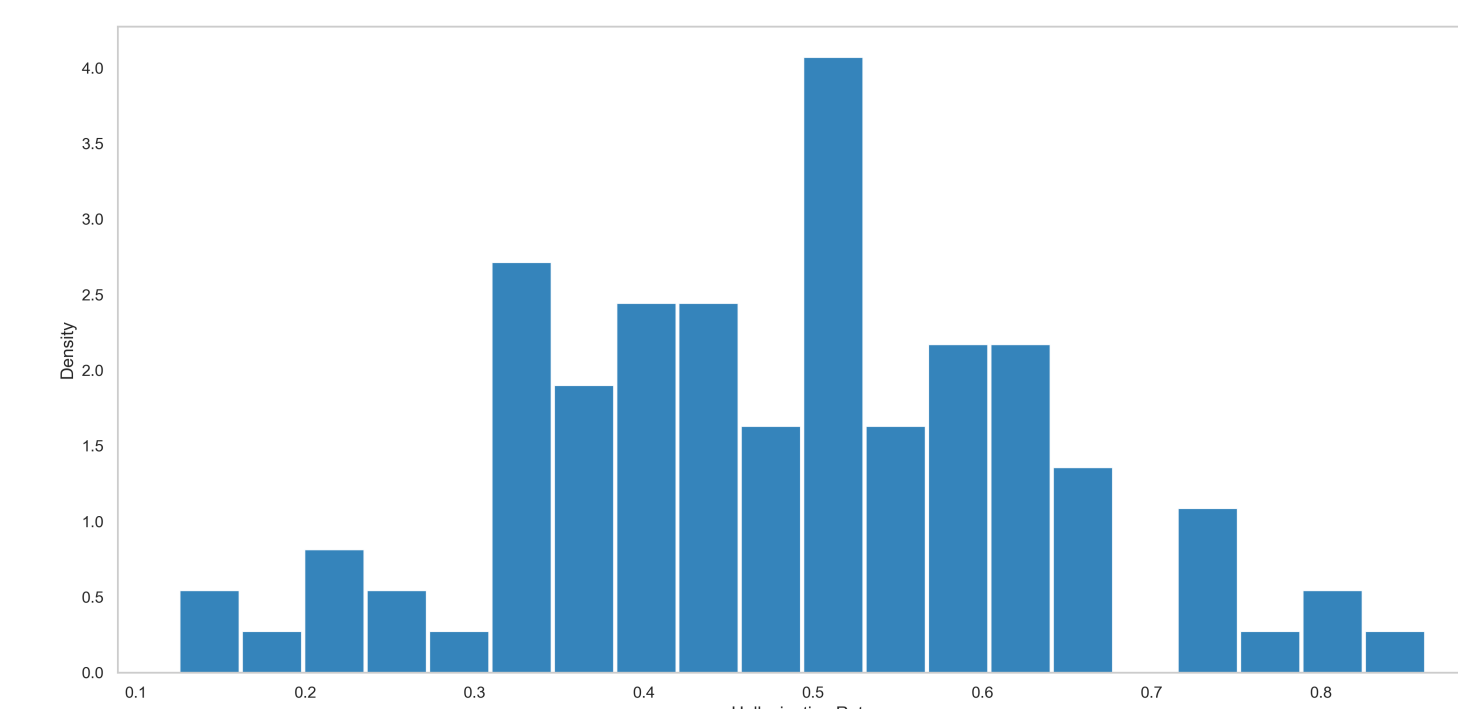


**Figure 1:** Correctness distribution.

## Conscious Incompetence

We remove the ground-truth evidence required to answer each question. A reliable model should recognize that the context is insufficient and abstain.

- Models frequently generate numerically specific claims or citations even when the necessary evidence is unavailable in the input context.



**Figure 2:** Hallucination rate when required evidence is removed.

Hallucination rate: [48.34%](#)

## Takeaways

- [Groundedness is strong](#) across modern citation-generation strategies.
- [Groundedness does not imply correctness](#).
- Citation precision is high, but [coverage remains incomplete](#).
- Missing evidence creates a major [hallucination](#) issue.

## Disclaimer

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